

Name : Satyajit Langote

Roll No : 41

Sub : FrontendForge

Class : MCA-I

Assignment- 5

Q.1 Student Result Component

Create a component **ResultCard**.

Display:

- Student Name
- Roll Number
- Marks
- Result (Pass/Fail)

Logic

- Marks $\geq 40 \rightarrow$ Pass
- Otherwise $\rightarrow$ Fail

Pass data using props

Use JavaScript expression inside JSX

Program :

```
import React from "react";
```

```
function ResultCard(props) {
```

```
    const result = props.marks >= 40 ? "Pass" : "Fail";
```

```
    return (
```

```
        <div>
```

```
            <h2>Student Name: {props.StudentName}</h2>
```

```
<h2>Roll Number: {props.RollNumber}</h2>
```

```
<h2>Marks: {props.marks}</h2>
```

```
<h2>Result: {result}</h2>
```

```
</div>
```

```
);
```

```
}
```

```
function App() {
```

```
  return (
```

```
    <div>
```

```
      <ResultCard
```

```
        StudentName="Satyajit"
```

```
        RollNumber="101"
```

```
        marks={75}
```

```
      />
```

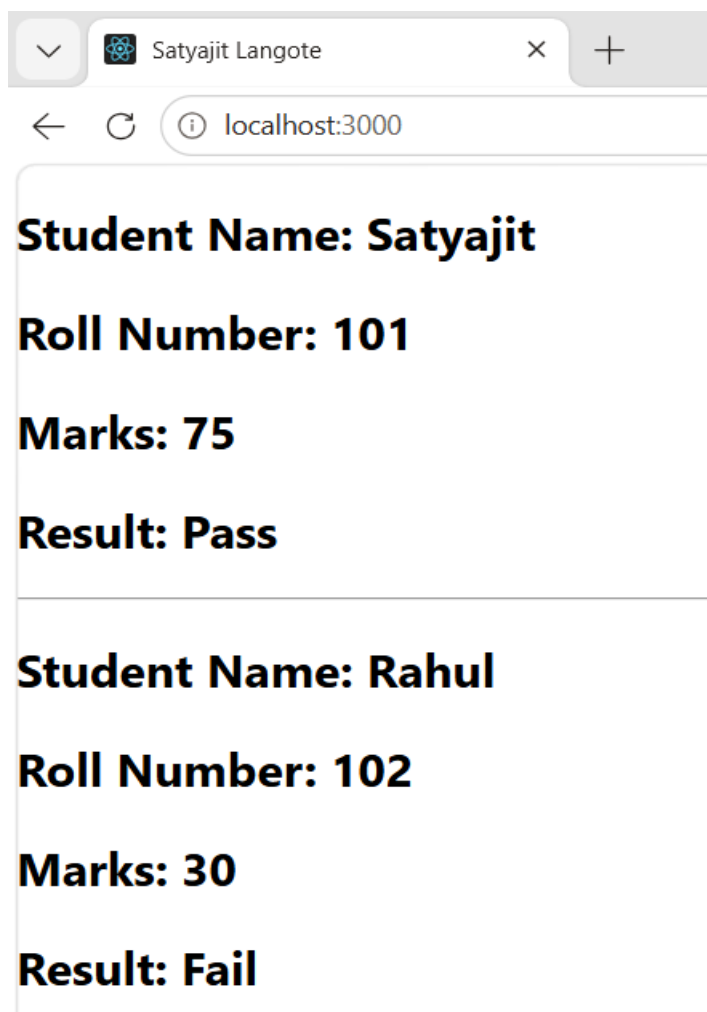
```
    <hr />
```

```
    <ResultCard
```

```
      StudentName="Rahul"
```

```
RollNumber="102"  
marks={30}  
/>  
  
</div>  
  
);  
}  
  
export default App;
```

Output :



Q.2 Grade Calculator

Modify previous program to also display **Grade**.

Marks	Grade
≥ 75	A
60–74	B
50–59	C
< 50	D

Use nested conditional expressions in JSX

Program :

```
import React from "react";

function GradeCalculator(props) {

  const result = props.marks >= 40 ? "Pass" : "Fail";

  const grade =

    props.marks >= 75 ? "A" :

    props.marks >= 60 ? "B" :

    props.marks >= 50 ? "C" : "D";

  return (

    <div>

      <h2>Student Name: {props.StudentName}</h2>

      <h2>Roll Number: {props.RollNumber}</h2>

      <h2>Marks: {props.marks}</h2>

      <h2>Result: {result}</h2>

      <h2>Grade: {grade}</h2>

      <hr />

    </div>
```

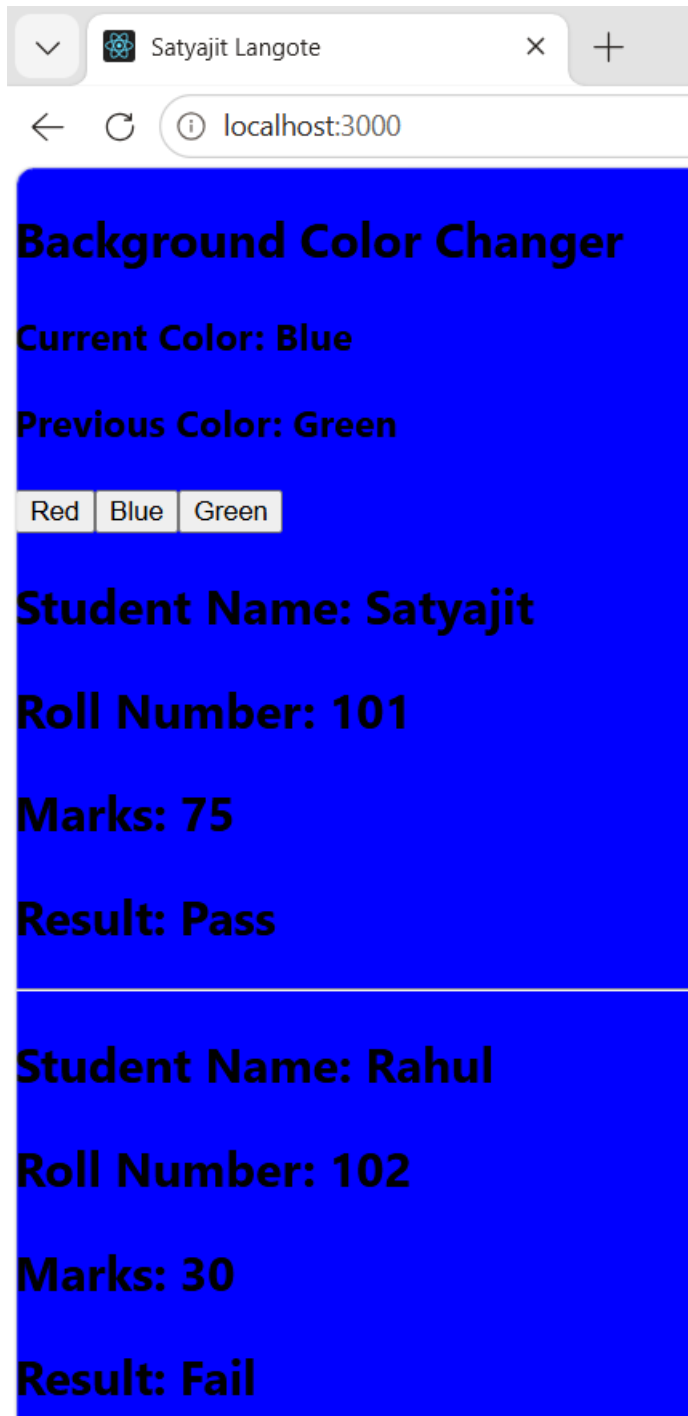
```
);  
}  
function App() {  
  return (  
    <div>  
      <GradeCalculator  
        StudentName="Satyajit"  
        RollNumber="101"  
        marks={85}  
      />  
      <GradeCalculator  
        StudentName="Rahul"  
        RollNumber="102"  
        marks={65}  
      />  
      <GradeCalculator  
        StudentName="Amit"  
        RollNumber="103"  
        marks={52}  
      />  
      <GradeCalculator  
        StudentName="Rohan"  
        RollNumber="104"  
        marks={35}  
      />  
    </div>  
  );  
}
```

```
    </div>

  );
}

export default App;
```

Output :



Q.3 Dynamic Fee Status

Create component **FeeStatus**

Display:

- Student Name
- Total Fee
- Paid Amount
- Remaining Amount

Logic

If remaining = 0 → "Paid"

Else → "Pending"

Calculate values using JavaScript inside JSX

Program :

```
import React from "react";
```

```
function FeeStatus(props) {  
  
    const remaining = props.totalFee - props.paidAmount;  
  
    const status = remaining === 0 ? "Paid" : "Pending";  
  
    return (  
        <div>  
            <h2>Student Name: {props.studentName}</h2>  
  
            <h2>Total Fee: {props.totalFee}</h2>  
  
            <h2>Paid Amount: {props.paidAmount}</h2>  
  
            <h2>Remaining Amount: {remaining}</h2>  
  
            <h2>Status: {status}</h2>  
  
            <hr />  
  
        </div>  
  
    );  
}
```

```
function App() {  
  return (  
    <div>  
      <FeeStatus  
        studentName="Satyajit"  
        totalFee={50000}  
        paidAmount={50000}  
      />  
      <FeeStatus  
        studentName="Rahul"  
        totalFee={50000}  
        paidAmount={30000}  
      />  
    </div>  
  );  
}  
export default App;
```

Output :

Student Name: Satyajit

Total Fee: 50000

Paid Amount: 50000

Remaining Amount: 0

Status: Paid

Student Name: Rahul

Total Fee: 50000

Paid Amount: 30000

Remaining Amount: 20000

Status: Pending

Q.4 Profile Card

Create component **ProfileCard**

Display:

- Name
- Course
- City
- Experience (years)
- Status

Logic

Experience $\geq 2 \rightarrow$ "Experienced"

Else $\rightarrow$ "Fresher"

Use props

Reuse component at least 3 times

Program :

```
import React from "react";
```

```
function ProfileCard(props) {
```

```
    const status=
```

```
    props.experience >= 2 ? "Senior" : "Junior";
```

```
    return (
```

```
        <div>
```

```
            <h2> Name : {props.name}</h2>
```

```
            <h2> Course : {props.course}</h2>
```

```
            <h2> Experience : {props.experience} years</h2>
```

```
            <h2> Status : {status} </h2>
```

```
        </div>
```

```
    );
```

```
}
```

```
function App() {
```

```
return (  
  <div>  
    <ProfileCard  
      name="Satyajit Langote"  
      course="Full-Stack-Java"  
      city="Sangli"  
      exprience={3}  
    />  
    <ProfileCard  
      name="Pradnya Mali"  
      course="B-Pharmacy"  
      City="Sangli"  
      exprience={2}  
    />  
    <profileCard  
      name="Rahul Patil"  
      course="JAva"  
      exprience={2}  
    />  
  </div>  
);  
}
```

export default App;

Output :

Name : Satyajit Langote

Course : Full-Stack-Java

Experience : years

Status : Junior

Name : Pradnya Mali

Course : B-Pharmacy

Experience : years

Status : Junior

Q5: Electricity Bill Calculation (Based on Units)

Create a program to calculate the electricity bill based on the number of units consumed.

Input:

- Consumer Name
- Units Consumed

Billing Rules (Slab System):

Units Range	Rate per Unit
0 – 100	₹5 per unit
101 – 300	₹7 per unit
Above 300	₹10 per unit

Output:

- Consumer Name
- Units Consumed
- Total Bill Amount

Program :

```
import React from "react";

function ElectricityBill(props) {

  let billAmount = 0;

  if (props.units <= 100) {

    billAmount = props.units * 5;

  }

  else if (props.units > 300) {

    billAmount = props.units * 7;

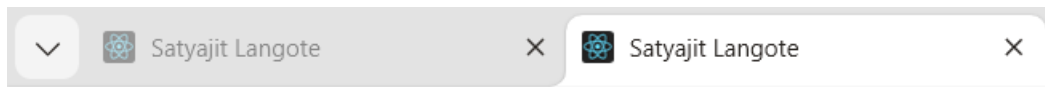
  }

  else {

    billAmount = props.units * 10;
```

```
}  
  
return (  
  <div>  
    <h2>Customer Name : {props.name}</h2>  
    <h2>Units Consumed : {props.units}</h2>  
    <h2>Bill Amount : {billAmount}</h2>  
  
  </div>  
  
);  
  
}  
  
function App(){  
  return (  
    <div>  
      <ElectricityBill name="Satyajit Langote " units={350}/>  
      <ElectricityBill name="Pradnya Mali" units={250}/>  
  
    </div>  
  
  );  
  
}  
  
export default App;
```

Output :



← ↻ ⓘ localhost:3000

Customer Name :Satyajit Langote

Units Consumed :350

Bill Amount : 2450

Customer Name :Pradnya Mali

Units Consumed :250

Bill Amount : 2500

P